

AI in Retail

Transforming the Shopping Experience

ITAI 2372 Mod -





Agenda

Overview of AI in Retail

Core AI Technologies in Retail

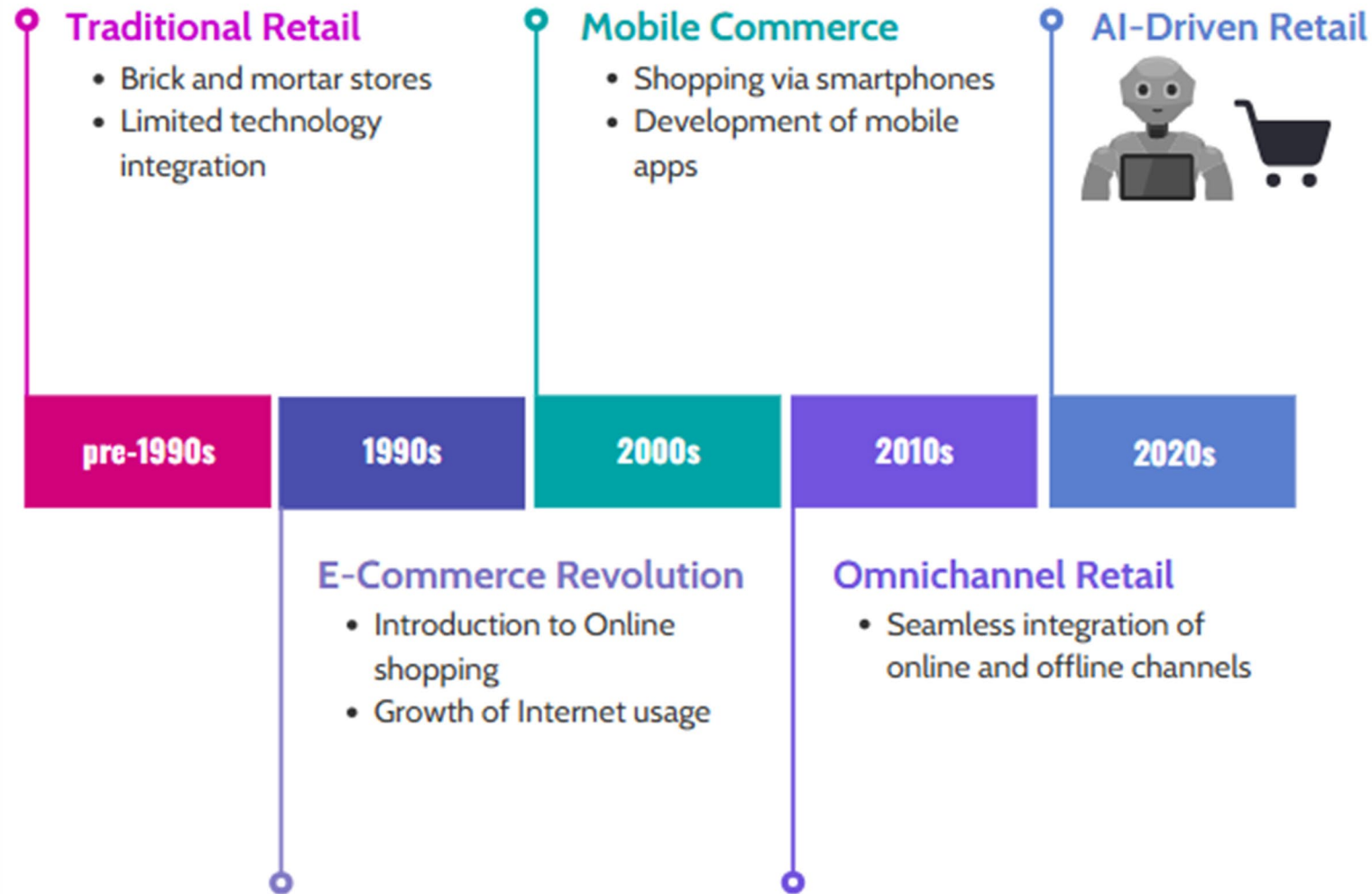
Applications of AI in Customer Experience

Supply Chain AI and Warehouse Automation

AI for Store Operations

Implementation Framework and Common Challenges

The Retail Technology Evolution





Key Drivers

- **Consumer Expectations**

- Personalization
- Convenience
- Speed

- **Business Pressures**

- Competition
- Margins
- Efficiency

- **Technology Evolution**

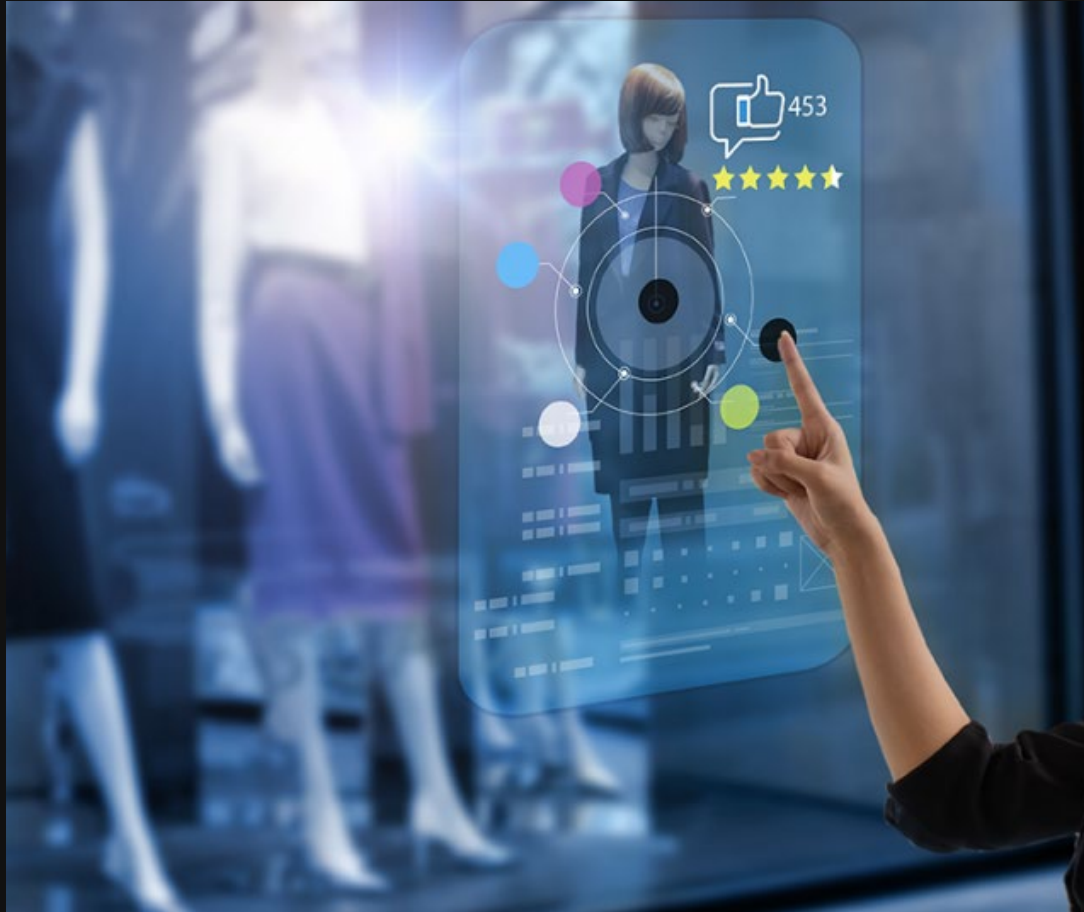
- Computing power
- Data availability

Core AI technologies (Review)

- Machine Learning
- Computer Vision
- Natural Language Processing
- Robotics
- Predictive Analytics



Computer Vision in Retail



- Applications of Computer Vision in Retail
 - Shelf monitoring
 - Customer tracking
 - Self-checkout systems
 - Loss prevention
 - Visual search

Natural Language Processing

- Customer Service Chatbots
- Voice Shopping
- Social Media Monitoring
- Customer Feedback Analysis
- Product Search Optimization





Personalization Engine

Data Collection

- Browse history
- Purchase history
- Demographics
- Context

Analysis

- Pattern recognition
- Preference learning
- Behavioral modeling

Output

- Product recommendations
- Custom messaging
- Personalized pricing
- Tailored experiences

Virtual and Augmented Reality



- Virtual Try-On
- Store Navigation
- Product Visualization
- Interactive Displays
- Training Applications

Smart Store Technologies

- Smart Shelves
- Interactive Mirrors
- Electronic Shelf Labels
- Automated Checkout
- Traffic Analysis



Supply Chain AI

- Demand Forecasting
- Inventory Optimization
- Route Optimization
- Supplier Management
- Risk Prediction



Warehouse Automation

Technologies

- Picking Robots
- Automated Guided Vehicles (AGVs)
- Sorting Systems
- Inventory Drones
- Smart Storage

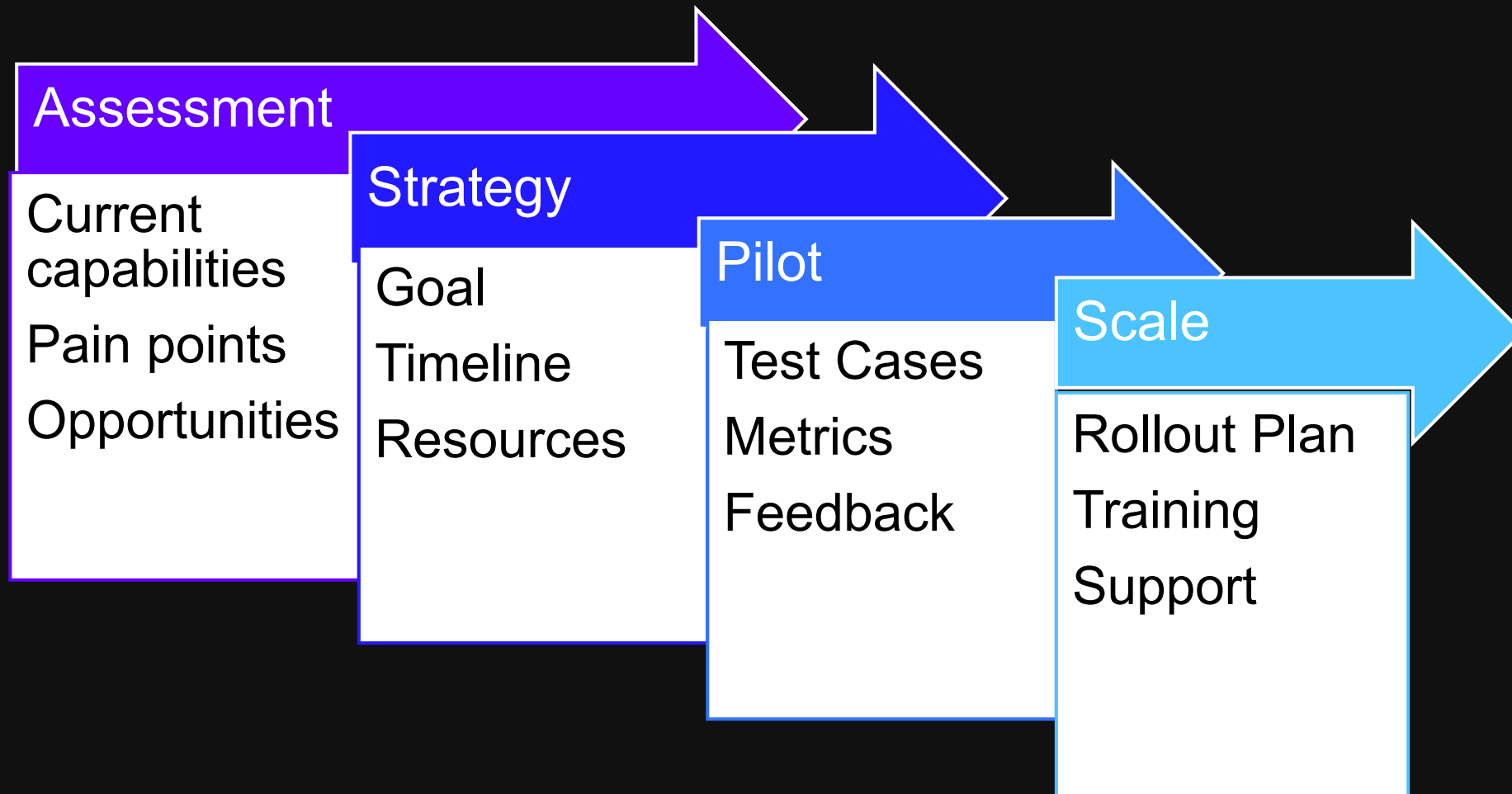


AI For Store Operations



- Staff Scheduling
- Maintenance Prediction
- Energy Optimization
- Security Systems
- Cleaning Automation

Implementation Framework



Common Challenges

- Data Quality Issues
- Integration Difficulties
- Change Management
- ROI Measurement
- Skill Gaps



Success Metrics



- KPIs
 - Customer satisfaction
 - Operational efficiency
 - Revenue impact
 - Cost reduction
 - Employee adoption



Ethical Considerations



Privacy

Concerns about customer data privacy



Bias

Potential for biased algorithms affecting decisions



Employment Impact

AI's effect on job displacement and creation



Data Security

Ensuring the security of sensitive data

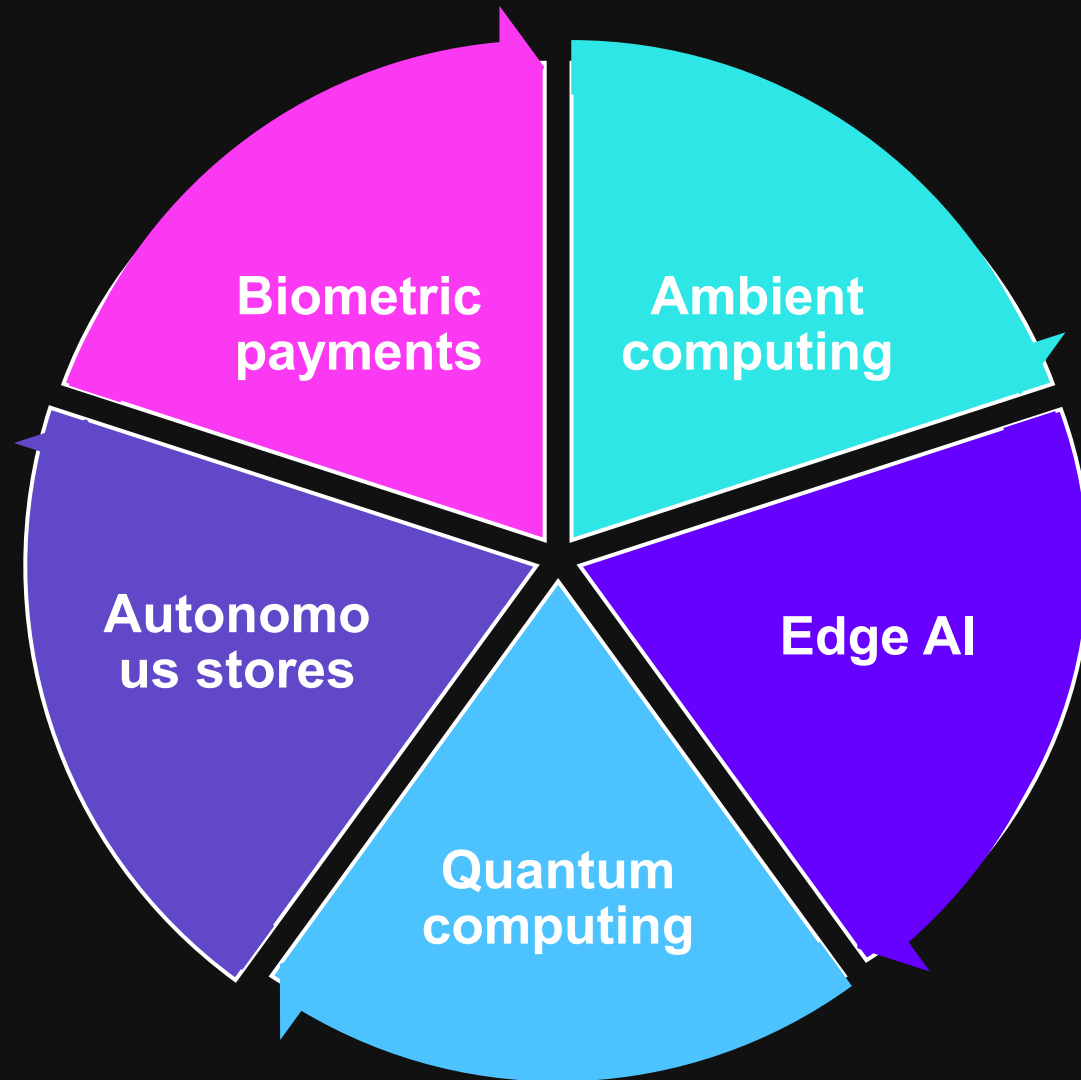


Transparency

Need for clear and understandable AI processes



Future Trends



Key Takeaways

- AI technologies are transforming the retail industry, from customer experience to supply chain management.
- Retailers must keep up with evolving technology to meet consumers' high expectations for personalization and convenience.
- Implementing AI in retail requires careful planning and consideration of unique challenges but can yield significant rewards.
- Successful AI implementation requires a focus on ethical considerations such as privacy, transparency, and bias.